Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5 HT	(a)			Total resistance = 8 + 4 = 12 (1) $I = \frac{12}{12\ (ecf)}$ (1) = 1 A (1)		3		3	3	
	(b)	(i)		$\frac{1}{R_1} = \frac{1}{12} + \frac{1}{8} = \frac{5}{24}$, so R_1 = 4.8 (1) $\frac{1}{R_2} = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$, so R_2 = 2.4 (1) total resistance = 4.8 + 2.4 = (7.20 [Ω]) (subs) (1) (ecf) (ecf)	1	2		3	3	
		(ii)		Current = $\frac{12}{7.2}$ = 1.67 (1) (allow ecf for 7.2) Current splits $\frac{2}{3}$ (1) Power = $\frac{4}{9} \times 12$ = 5.33 W (1) **Alternative** Current = $\frac{12}{7.2}$ = 1.67 (1) (allow ecf for 7.2) Pd across first parallel 1.67 (ecf) × 4.8 (ecf) = 8.016 I in 12 Ω resistor is $\frac{8.016}{12}$ (ecf) = 0.668 A (1) P = I^2R = 5.35 [W] (1)		3		3	3	
				Question 5 total	**1**	**8**	**0**	**9**	**9**	**0**